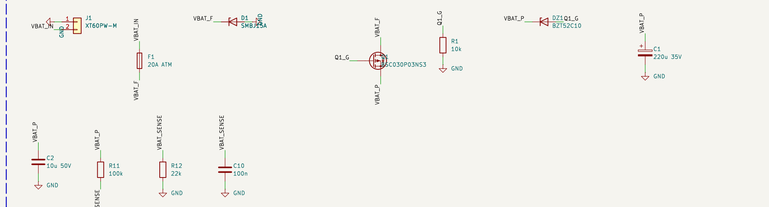
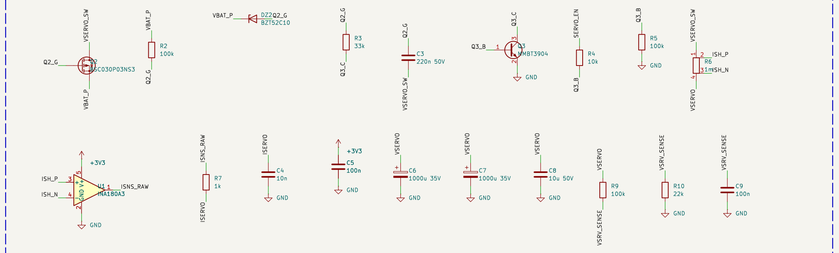
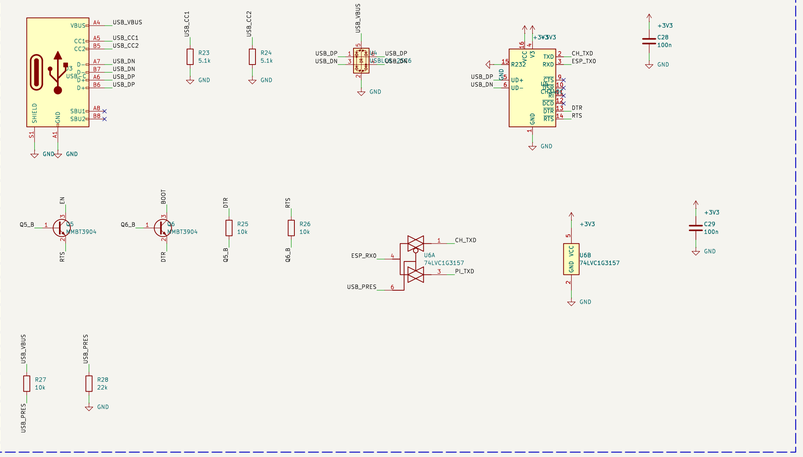
Battery input: XT60, 20 A fuse, TVS, reverse-polarity P-FET



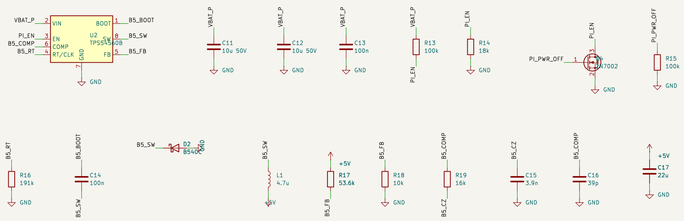
Servo rail: ESP32-switched P-FET (soft start), current shunt, bulk



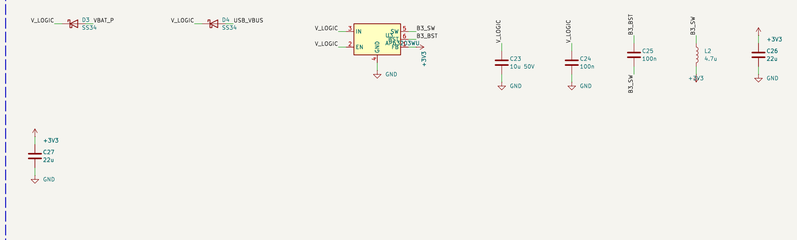
USB-C, CH340C, auto-reset, UART0 source mux (USB wins)



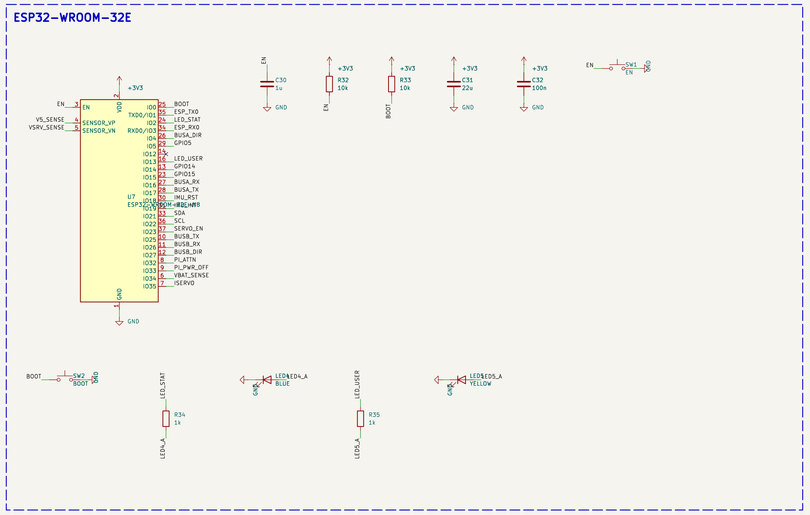
5 V / 5 A buck for the Raspberry Pi 5 (TPS54560B)



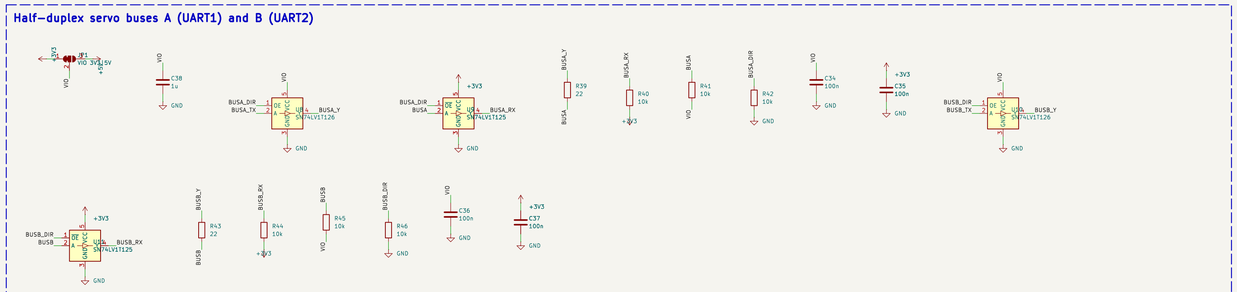
3.3 V logic buck (diode-OR battery / USB)



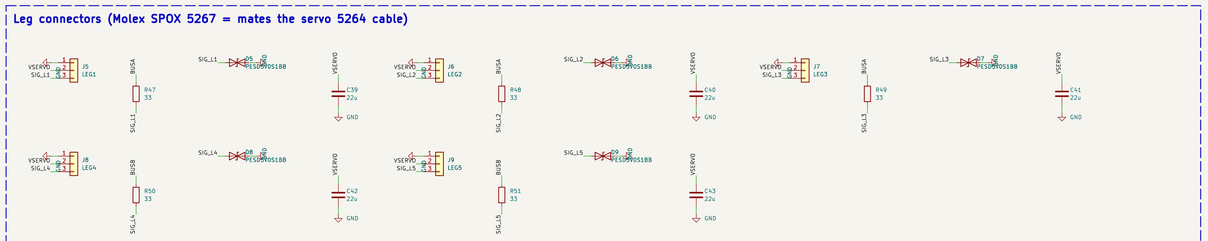
ESP32-WROOM-32E



Half-duplex servo buses A (UART1) and B (UART2)



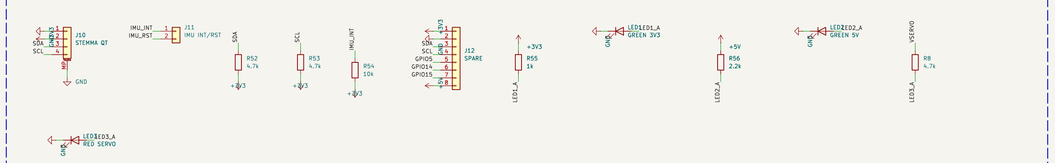
Leg connectors (Molex SPOX 5267 = mates the servo 5264 cable)



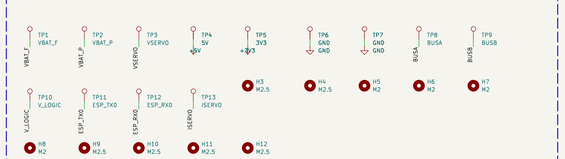
Raspberry Pi 5 link (IDC 2x5 to Pi header pins 5–14)



IMU (STEMMA QT), spare header, LEDs, sense dividers



Test points and mounting holes



Power flags (ERC: these nets are driven by regulators through passives)

Generated by tools/gen_sch.py from tools/design.py

ESP32-WROOM-32E 2x half-duplex servo bus + 20 A battery input + 5 V/5 A Pi supply

Matthew Lin

Sheet: /

File: rocky_ctrl.kicad_sch

Title: Rocky pentapod control board

Size: A1 Date: 2026-10-19 **Rev: v1**

KiCad 9.0.5

Rocky control board v1 — see DESIGN.md for the current budget, pin map and placement.